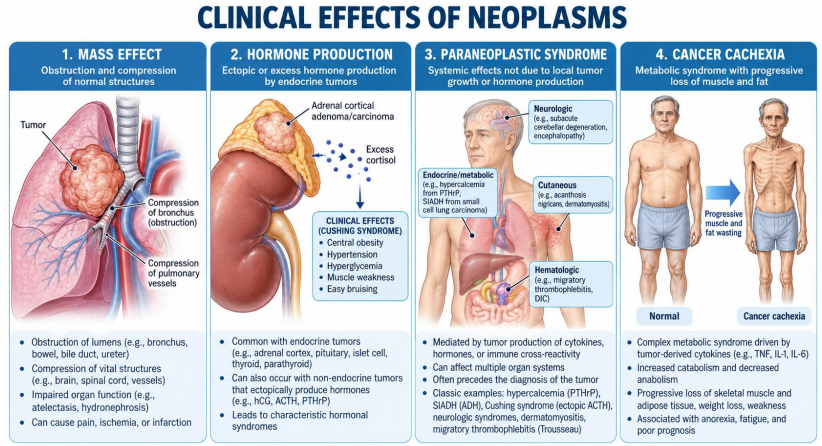
Clinical aspects, grading and staging of tumors

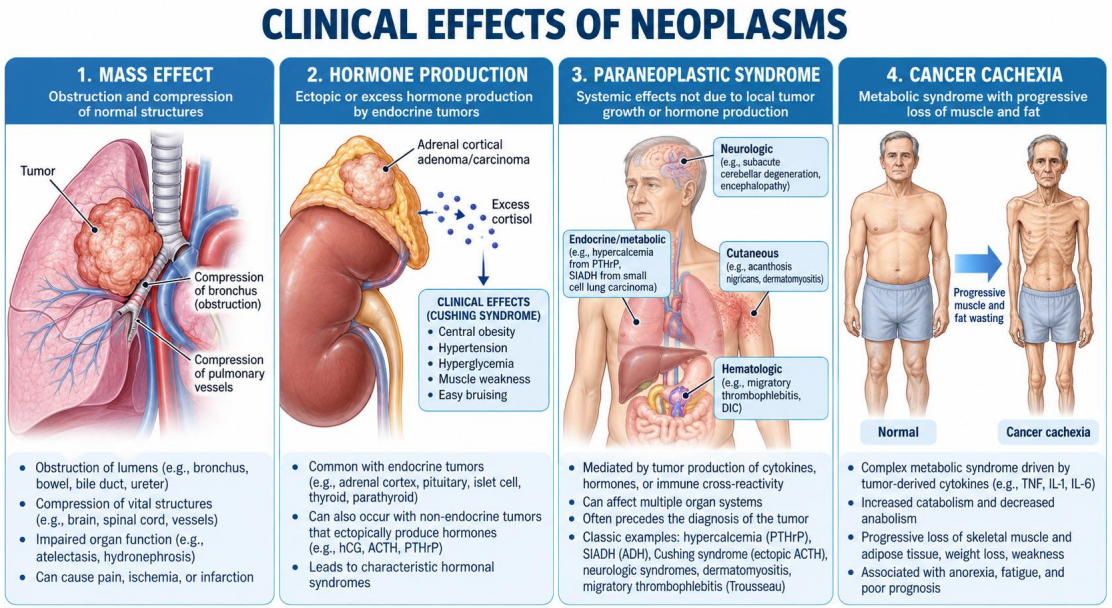
(SGL10)

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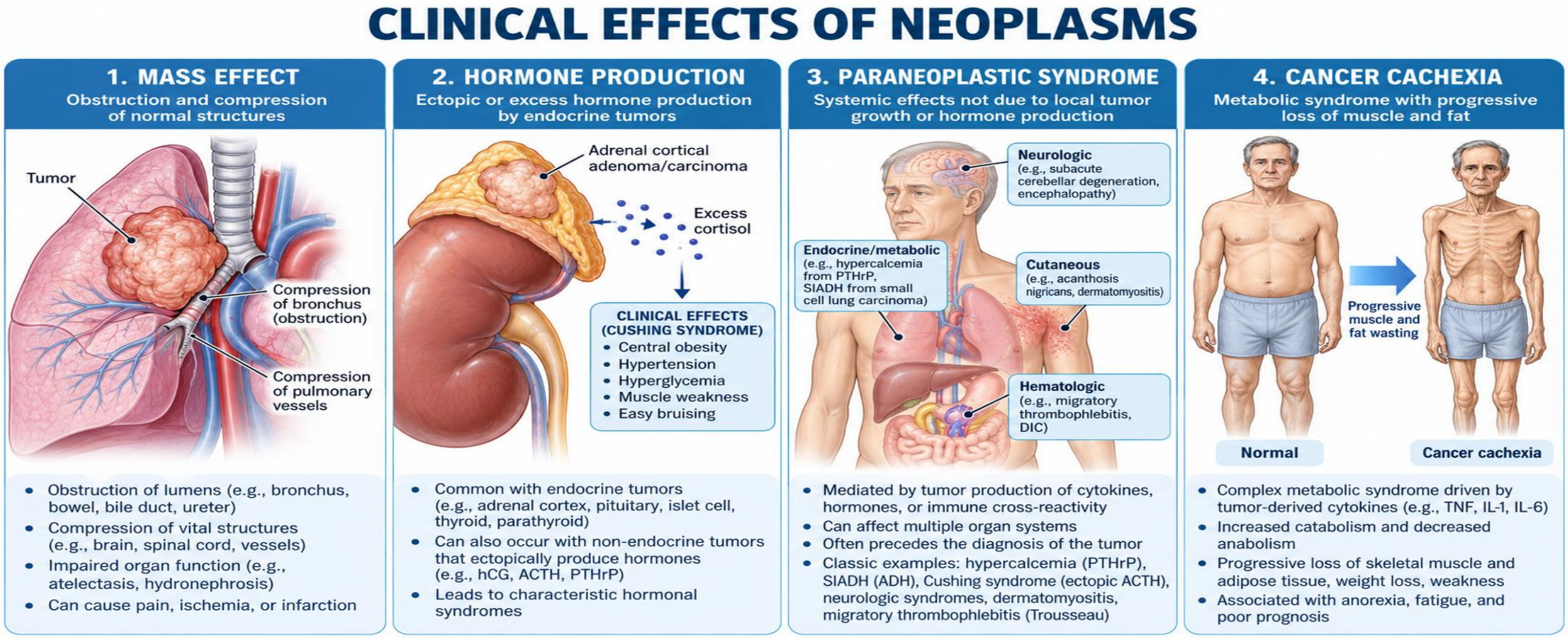
Lecture outlines:

- Clinical effects of benign and malignant neoplasms
- Mass effect, hormone production, paraneoplastic syndromes and cachexia
- Cancer grading and TNM staging
- Laboratory diagnosis: biopsy, cytology, IHC, molecular tests and tumor markers



Clinical effects of neoplasms

- Most tumours are asymptomatic until they are at an advanced stage.
- The clinical features can be diverse and complex depending on the site and size of the tumour.

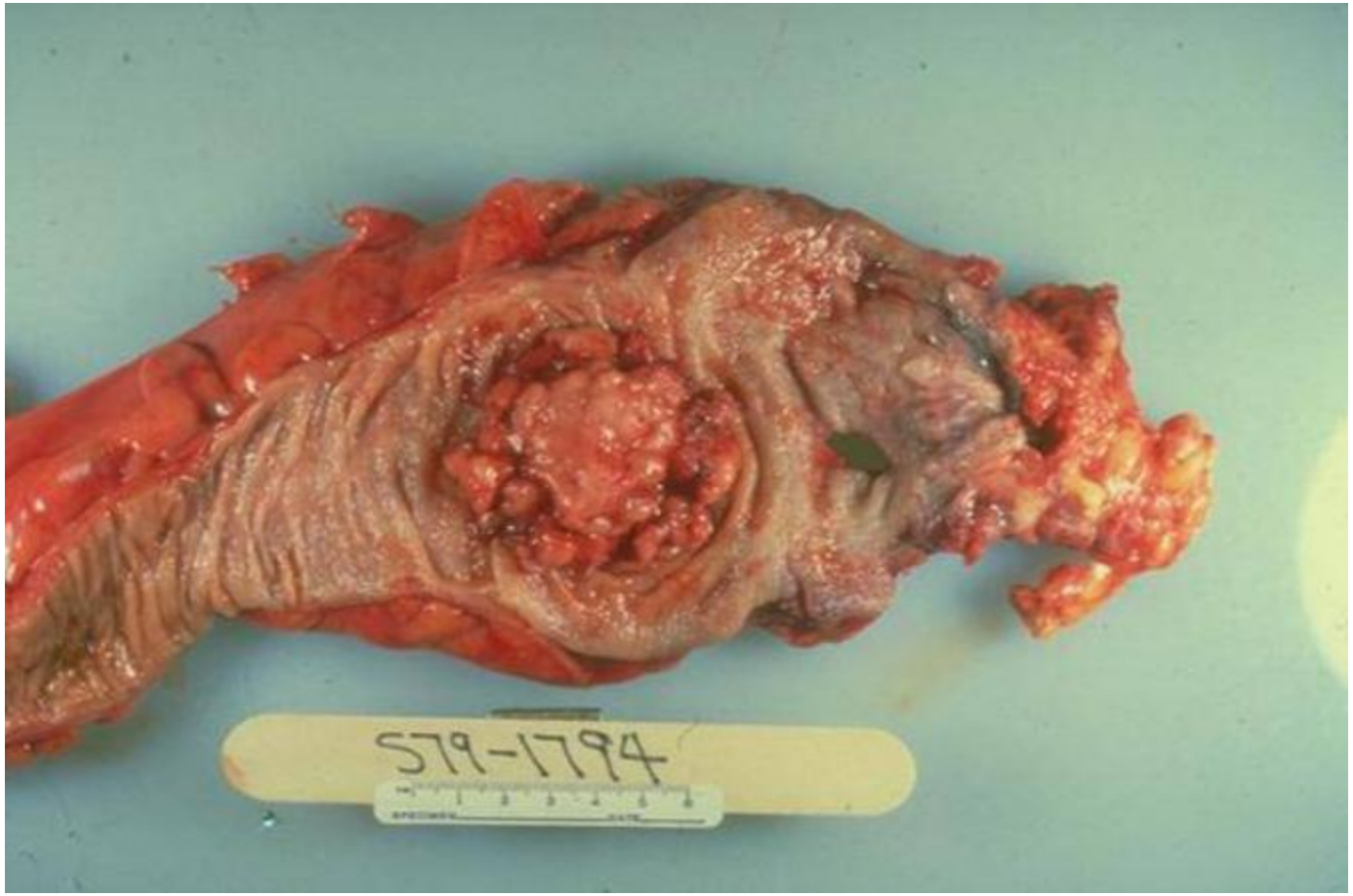


1. Mass effect

- Many tumors are asymptomatic until advanced.
- A lump may cause pressure, obstruction, ulceration or bleeding.
- Clinical manifestations depend strongly on site and size.
- Examples:; colon cancer causing obstruction or bleeding; brain tumor causing seizures or neurologic dysfunction.



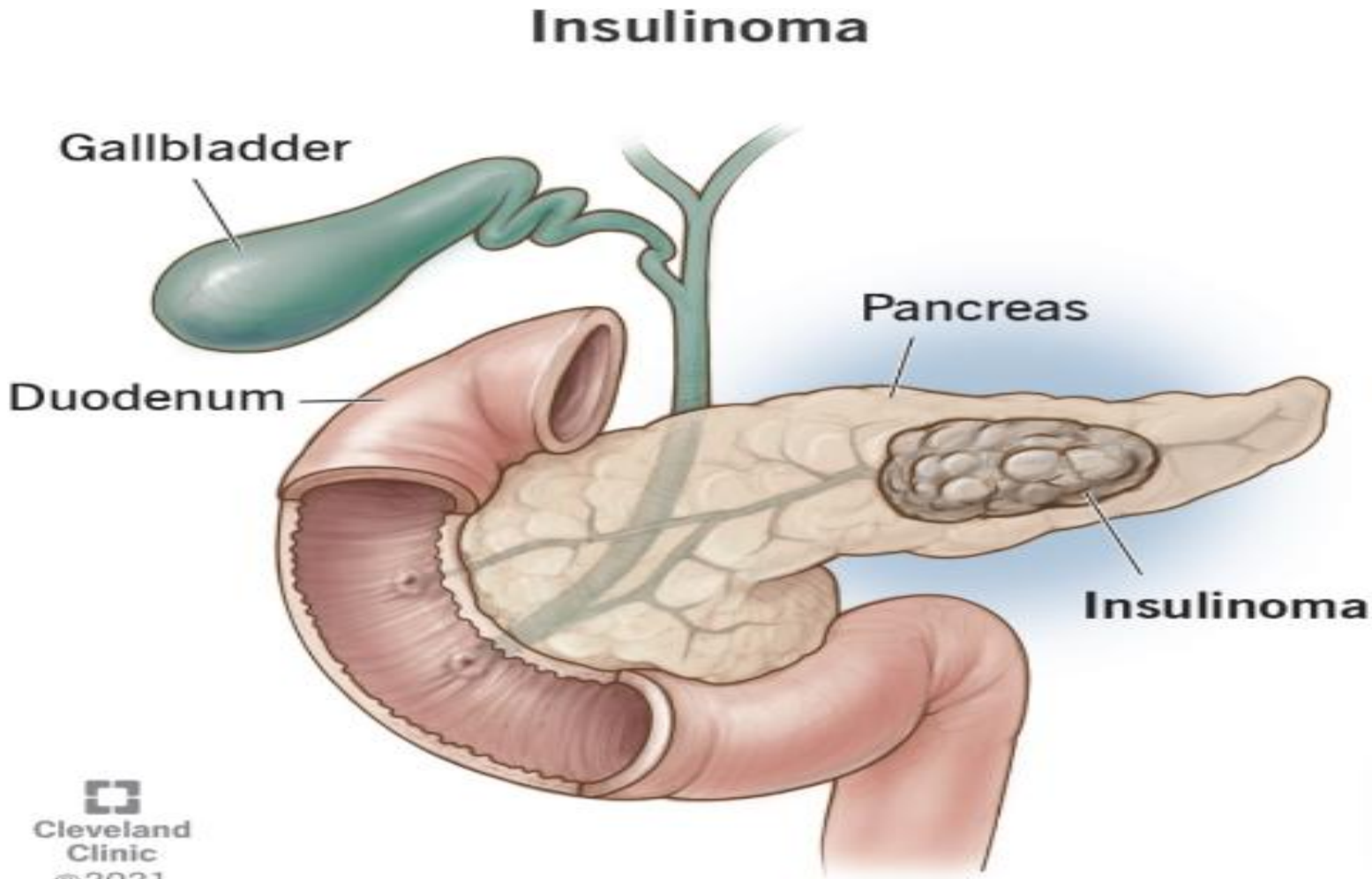
A small (1-cm) pituitary adenoma can compress and destroy the surrounding normal gland, giving rise to hypo pituitarism.



Adenocarcinoma of colon. This is an opened colon shows an ulcerated carcinoma with elevated, heaped-up edges that may cause bleeding.

2. Hormone production

Tumor example	Hormone / product	Clinical effect
Pituitary adenoma	ACTH	Cushing syndrome
Pancreatic insulinoma	Insulin	Hypoglycemia
Parathyroid adenoma	Parathyroid hormone	Hypercalcemia
Some non-endocrine tumors	Ectopic hormones	Paraneoplastic endocrine syndromes


Insulinoma
Gallbladder
Pancreas
Duodenum
Insulinoma
Cleveland Clinic
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3. Paraneoplastic syndromes

- Systemic symptoms not explained by local tumor spread or by hormones normally produced by the tissue of origin.
- May be caused by tumor-derived hormones, cytokines or immune cross-reactivity.
- Common abnormalities include hypercalcemia and ectopic ACTH production by small-cell lung carcinoma.
- Other examples include clubbing/hypertrophic osteoarthropathy in lung cancer.

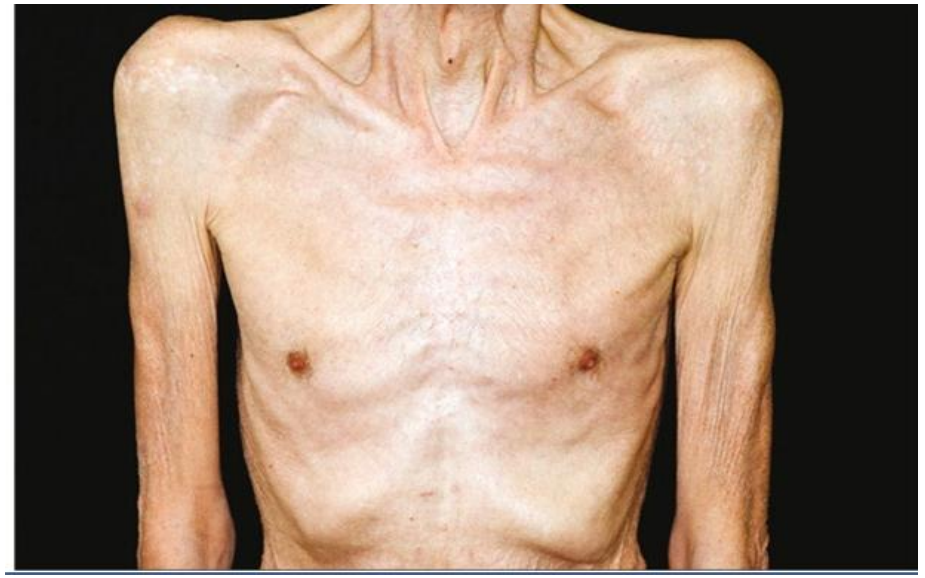
-A clinical sign of internal malignancy (peripheral non-small cell ca of lung) or chronic disease.



-The underlying disease usually precedes the cutaneous sign, however, there have been cases of the clubbing preceding identification of internal malignancy.

4. Cancer cachexia

Feature	Key point
Definition	Progressive loss of body fat and skeletal muscle with weakness
Correlation	Often worse with larger and more widespread cancers
Mechanisms	Reduced appetite; metabolic changes; tumor/macrophage-derived cytokines such as TNF; proteolysis-inducing factors
Clinical importance	Cannot be fully corrected by nutrition alone and worsens functional status

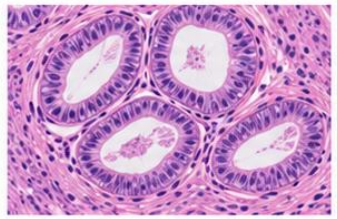


Grading and staging:

Histologic Grading of Cancer

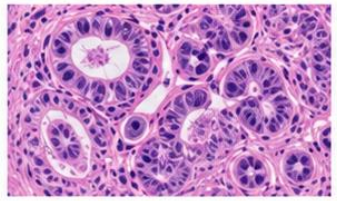
Based on degree of differentiation (how much tumor cells resemble normal cells)

Robbins: Grading reflects tumor dif erentiation, which correlates with biologic behavior and prognosis.



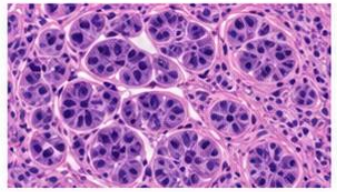
Grade 1 (Well differentiatted)

- Tumor cells closely resemble normal cells
- Preserved architecture (e.g., gland formation in adenocarcinoma)
- Less atypia, lower mitotic rate
- Generally better prognosis



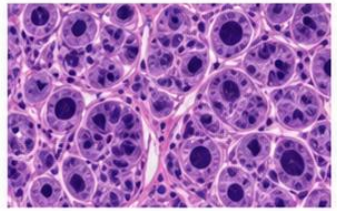
Grade 2 (Moderately differentiated)

- Intermediate resemblance to normal cells
- Partial loss of normal architecture
- Moderate nuclear atypia and mitotic activity
- Intermediate prognosis



Grade 3 (Poorly differentiated)

- Little resemblance to normal cells
- Loss of normal architecture
- Marked nuclear atypia, increased mitoses
- More aggressive behavior, worse prognosis



Grade 4 (Undifferentiated)

- No resemblance to normal cells
- No recognizable architecture
- Marked atypia, often with numerous mitoses and tumor giant cells
- Worst prognosis

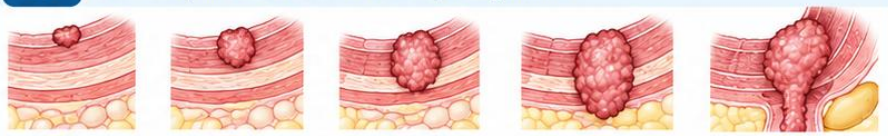
TNM Staging of Cancer

Based on anatomic extent of disease

Robbins: The TNM system classifies the anatomic extent of cancer and is the basis for overall stage grouping (Stage I–IV), which correlates with prognosis and guides therapy.

T — Primary tumor

Size and/or local extension of the primary tumor



Tis — Carcinoma *in situ*

T1 — Invades submucosa

T2 — Invades muscularis propria

T3 — Invades through muscularis propria into pericolonic tissue

T4 — Invades adjacent organs or structures

N — Regional lymph nodes

Number of involved regional (usually draining) lymph nodes



N0 — No regional lymph node metastasis

N1 — Metastasis in 1–3 regional lymph nodes

N2 — Metastasis in 4 or more regional lymph nodes

N3 — Metastasis in regional lymph node(s) with extracapsular extension (matted nodes)

M — Distant metastasis

Spread to nonregional distant sites



M0 — No distant metastasis

M1 — Distant metastasis present (e.g., lung, liver, bone, brain, etc.)

Cancer grading

- Grading is a histologic assessment performed by the pathologist.
- It is based mainly on degree of tumor-cell differentiation and cytologic atypia.
- It predicts biologic aggressiveness: poorly differentiated tumors usually behave more aggressively.

Cancer staging

Component	Meaning	Assessment
T	Primary tumor size and local anatomic extension	Clinical, radiologic and pathologic data
N	Regional lymph-node involvement	Examination, imaging, sentinel node or lymph-node dissection
M	Presence or absence of distant metastasis	Imaging, biopsy/cytology when needed
Why stage matters	More powerful predictor of prognosis than grade	Guides treatment selection

Who determines stage?

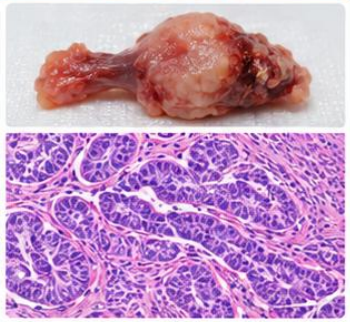
Type of staging	Main source of information
Clinical staging	History, physical examination, endoscopy, radiology and biopsy before treatment
Radiologic staging	Imaging such as CT, MRI, ultrasound and PET scan
Pathologic staging	Surgical specimen and lymph nodes examined by pathologist
Integrated stage	Combines available data to guide treatment and prognosis

Laboratory diagnosis of cancer

LABORATORY DIAGNOSIS OF CANCER

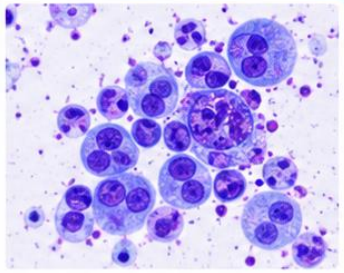
COMPLEMENTARY TESTS PROVIDE A COMPREHENSIVE UNDERSTANDING OF THE TUMOR

BIOPSY SPECIMEN
HISTOPATHOLOGY



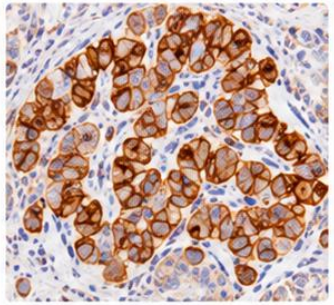
- Tissue obtained by core needle, endoscopic or surgical biopsy
- Examined by a pathologist
- Determines tumor type, grade and histologic features

CYTOLOGY SMEAR
FINE NEEDLE ASPIRATION (FNA) OR FLUID CYTOLOGY



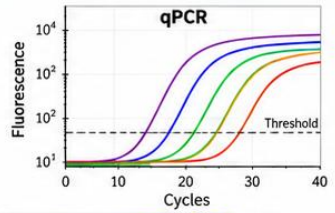
- Minimally invasive sampling
- Rapid evaluation
- Useful for many solid tumors (e.g., lung, thyroid, lymph node) and body fluids (e.g., pleural, ascitic fluid)

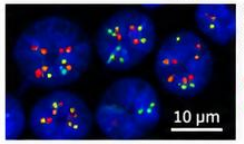
IMMUNOHISTOCHEMISTRY (IHC)



- Uses antibodies to detect tumor-specific proteins
- Helps confirm the diagnosis and identify tumor origin
- Provides predictive markers (e.g., ER, PR, HER2, PD-L1)

MOLECULAR TESTING
PCR, NGS, FISH





FISH
HER2 gene amplification
HER2
CEP17

- Detects genetic alterations (e.g., mutations, fusions, copy number changes)
- Guides targeted therapy
- Methods include PCR, next-generation sequencing (NGS) and FISH

TUMOR MARKER BLOOD TEST



PSA	Prostate cancer
CEA	Colorectal and other cancers
CA 125	Ovarian cancer
CA 19-9	Pancreatic and biliary cancers
AFP	Liver cancer (hepatocellular carcinoma, germ cell tumors)
β-hCG	Germ cell tumors
CA 15-3	Breast cancer

- Measured in blood
- Useful for screening (selected cancers), monitoring treatment response and detecting recurrence
- Not typically diagnostic on their own

INTEGRATED LABORATORY RESULTS

DIAGNOSIS
- Confirms malignancy
- Determines tumor type and origin
- Defines key pathologic and molecular features

PROGNOSIS
- Tumor grade and stage
- Molecular and protein markers
- Helps predict risk of recurrence and overall outcome

TREATMENT SELECTION
- Identifies actionable targets (e.g., HER2, EGFR, ALK, BRAF, MSI, PD-L1)
- Guides use of targeted therapy, immunotherapy and other personalized treatments
- Supports monitoring of treatment response

Sampling methods

Method	Use / advantage	Limitation
Excisional biopsy	Entire lesion is removed; architecture preserved	May not be feasible for large/deep lesions
Incisional / core biopsy	Representative tissue for histology and IHC	Requires good sampling and preservation
Fine-needle aspiration cytology	Rapid, minimally invasive; useful in breast, thyroid and lymph nodes	Less tissue architecture
Cytologic smears / brushings	Screening or surface lesions	May need follow-up tissue biopsy

Immunohistochemistry (IHC)

- Uses antibodies to detect specific antigens in tissue sections.
- Helps identify and classify undifferentiated tumors.
- Useful in categorizing leukemias and lymphomas.
- Can suggest the site of origin of metastases, e.g., PSA in prostate carcinoma.
- Provides prognostic and predictive markers for treatment decisions.

Molecular diagnosis

Technique	What it detects	Clinical value
PCR / qPCR	Specific mutations, infections or gene rearrangements	Diagnosis and minimal residual disease in selected tumors
FISH	Translocations, amplifications and copy-number changes	Examples: gene fusions, HER2 amplification
Next-generation sequencing	Multiple mutations across cancer genes	Classification, prognosis and targeted therapy selection
Personalized medicine	Genetic makeup of tumor	Guides targeted therapy and immunotherapy choices

Tumor markers

Marker	Common association	Best use
CA-125	Ovarian cancer	Monitoring and recurrence assessment
AFP	Hepatocellular carcinoma; testicular germ-cell tumors	Diagnosis support and monitoring
CEA	Colorectal carcinoma	Follow-up and recurrence monitoring
PSA	Prostatic carcinoma	Screening context and monitoring
CA 19-9	Pancreatic and gastric cancers	Monitoring, not primary diagnosis alone
hCG	Choriocarcinoma / germ-cell tumors	Diagnosis support and monitoring

Summary

- Tumors affect patients by mass effect, hormone production, paraneoplastic syndromes and cachexia.
- Grade describes histologic differentiation; stage describes anatomic extent of disease.
- Stage is usually the stronger predictor of prognosis and treatment planning.
- Good-quality biopsy with clinical details remains central to cancer diagnosis.
- IHC, molecular testing and tumor markers refine diagnosis, prognosis and treatment selection.

Thank you

Questions?